

SafeDig AI — Technology Stack Specification

Classification: Enterprise AI & Safety-Critical Engineering

Standard: UK Health and Safety Guidance 47 (HSG47) & CDM 2015

Core Invariant: Zero Escaped Hazards (`SAFE_MODE=True`)

1. Architectural Philosophy: Dual-Engine Verification

SafeDig uses a segregated dual-engine model specifically engineered for zero-hallucination utility clearance:

1. Deterministic Geometry & Computer Vision Core:
- PyMuPDF vector math, Shapely 2D spatial geometry, OpenCV image analysis, and a 17-Gate Policy Engine.
 - 100% deterministic, auditable, bit-for-bit reproducible, zero hallucination.
2. AI / LLM Advisory Layer:
- LangGraph stateful DAG orchestration paired with local on-premise LLMs (Qwen 2.5 / Llama 3.2).
 - Generates natural-language summaries, discrepancy explanations, and operator advisory notes.
 - Zero Validation Authority:** Cannot override, clear, or relax any safety gate.

2. Technology Stack Breakdown

Backend & Core Server Framework

Component	Technology	Version	Role in Platform
Language	Python	>=3.11	High-speed typing, pattern matching, async core
REST API	FastAPI	>=0.100.0	Asynchronous REST endpoints, OpenAPI auto-docs
ASGI Server	Uvicorn	>=0.22.0	High-throughput non-blocking production runtime
Data Validation	Pydantic v2 & Settings	>=2.0.0	Strict domain models, environment config
Async Client	HTTPX & AnyIO	>=0.25.0	Non-blocking API calls and concurrency

Computer Vision, Spatial Geometry & Vector Extraction

Component	Technology	Version	Role in Platform
Vector Engine	PyMuPDF (fitz)	1.28.2	Vector path extraction, dash analysis, 300 DPI rasterizer
Spatial Geometry	Shapely	>=2.0.0	Exact 2D polygon intersection, AOI bounding & buffer math
Computer Vision	OpenCV (headless)	>=4.8.0	HSV color masking, contour detection, image crops
Spreadsheets	Pandas & OpenPyXL	>=2.0.0	Statutory index parsing (<code>index.xlsx</code>) & normalization
Database	SQLAlchemy + aiosqlite	>=2.0.0	Async SQLite (<code>safedig.db</code>) & PostgreSQL production support
Migrations	Alembic	>=1.13.0	Schema migration tracking and revision history
PDF Reporting	ReportLab	5.0.1	Automated generation of audit reports and compliance specs

AI, Local LLMs & Orchestration

Component	Technology	Role in Platform
Pipeline DAG	LangGraph (>=0.2.0)	Stateful deterministic pipeline orchestration
Core Abstraction	LangChain-Core (>=0.3.0)	Standardized messages, prompts & agent states
Primary LLM	Qwen 2.5 (7B / 14B) via Ollama	Technical reasoning, discrepancy explanations
Edge LLM	Llama 3.2 (3B) via Ollama	Lightweight low-resource edge deployments
Deterministic Fallback	Python Rule Explainer	Oms offline fallback if LLM is unavailable

Frontend Console & Inspection Engine

Component	Technology	Role in Platform
Framework	Vanilla JS + Tailwind CSS	Responsive, zero-dependency, high-speed single page console
Vector Icons	Lucide Icons	Visual status indicators, gate badges, control buttons
Canvas Engine	DOM Hardware-Accelerated Zoom	Pan (drag) & Zoom (50%–600%), AOI centering & radar scanner loader

3. Infrastructure & Deployment Environment

Tier	Technology	Specification / Configuration
Container Engine	Docker & Docker Compose	Multi-stage lightweight image on Python 3.11-slim base
Reverse Proxy	Nginx Alpine	SSL termination, client max body size 100MB, Gzip compression
Process Daemon	Linux Systemd	Persistent service management with automatic restart on failure
Database Storage	SQLite with WAL Mode	Zero-config, single-file forensic database with concurrent reads

4. Hardware Sizing & Recommended Profiles

Environment	CPU	RAM	Storage	Local LLM Profile
Local Dev Workstation	4 Cores (x86_64 / ARM64)	16 GB	20 GB NVMe	Qwen 2.5 (7B) via Ollama (GPU or CPU)
Field Excavator Van (Edge)	4 Cores (Intel N100 / Pi 5)	8 GB	64 GB SSD	Llama 3.2 (3B) or Oms Rule Fallback
Enterprise Central Server	16+ Cores	64 GB	500 GB NVMe	Qwen 2.5 (14B / 32B) with NVIDIA CUDA

5. Mandatory Safety Rule: Zero LLM Validation Authority

Under SafeDig enterprise policy:

- The LLM has **ADVISORY AUTHORITY ONLY**.
- No safety gate, auto-clear decision, or hazard finding can be overridden by model inferences.
- If the LLM produces a contradiction or becomes unresponsive, the system automatically defaults to deterministic safety rules with zero latency.